Record Renewal Compatibility in Collatz Synchronization

Paper N: Two-Gate Jump Targets, Local vs. Long-Range Obstructions, and the Arithmetic Memory Problem

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Abstract

This paper completes the two-gate jump framework for Collatz synchronization records and clarifies the exact scope of the local obstruction theory developed in Papers F–M.

Theorem JE2 (Jump from $a = 2$, proved): At a semi-renewal event $a_t = 2$ with $n_t = 1 + 4q_t$, letting $s = v_2(3q_t + 1) \geq 1$, the next value is $T(n_t) = (3q_t + 1)/2^s$ and:

$$a_{t+1} = v_2\left(\frac{3q_t + 1}{2^s} - 1\right).$$

Height K is achieved iff $q_t \equiv r_{K,s} \pmod{2^{K+s}}$, where $r_{K,s} = (2^s - 1) \cdot 3^{-1} \bmod 2^{K+s}$.

Scope correction for Papers J–M: The phase transport identity $q_{\tau_i} = 3^{S_i} q_{\tau_1}$ (used in Papers J–M) is exact within a single cascade (Theorem OC), but does NOT hold between distinct renewal events. Each renewal event resets q's residue class through the full Syracuse dynamics. Consequently, Papers J–M establish *local cascade-level* obstructions, not global record-to-record transport constraints.

The fundamental open question (Paper N): A divergent orbit requires an infinite sequence of renewal events $\tau_1 < \tau_2 < \cdots$ at which $q_{\tau_i} \equiv \text{target}_{K_i}$. Between τ_i and τ_{i+1}, q evolves through complex Syracuse dynamics. The question is whether these dynamics preserve or destroy the "arithmetic memory" of previous records—equivalently, whether $q_{\tau_{i+1}}$ is effectively independent of q_{τ_i} modulo $2^{K_{i+1}}$.

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1 Introduction

1.1 Two kinds of renewal events

The valuation process $a_t = v_2(n_t - 1)$ (Paper F) has two types of states where a high jump can be launched:

- *Gate 1* ($a_t = 1$): $n_t \equiv 3 \pmod 4$. Theorem JE (Paper H): $a_{t+1} = v_2(3q_t + 1)$, with height K iff $q_t \equiv r_K \pmod{2^K}$.
- *Gate 2* ($a_t = 2$): $n_t \equiv 5 \pmod 8$. The present paper establishes Theorem JE2, giving the parallel formula.

States with $a_t \geq 3$ undergo deterministic descent (Theorem VD, Paper F) and cannot launch new high records.

1.2 Scope clarification for Papers J–M

Papers J–M developed a phase transport model: between records, $q_{\tau_i} = 3^{S_i} q_{\tau_1}$. We show this identity is exact within a single cascade (Theorem OC, Paper G) but fails across distinct renewals. The phase theory of Papers J–M therefore applies at the *cascade level*: it correctly characterizes the immediate obstruction after a height-K jump, but does not directly constrain how q evolves between non-adjacent records.

1.3 The record chain problem

Any divergent Collatz orbit must generate an infinite strictly increasing sequence of jump heights (Theorem Monotonicity, Paper M). Each height record requires q to lie in a specific

thin residue class at the corresponding renewal event. The map $q_{\tau_i} \mapsto q_{\tau_{i+1}}$ is complex Syracuse dynamics. Paper N identifies this "arithmetic memory" problem as the core open question.

2 Theorem JE2: Jump Formula from Gate 2

Theorem 2.1 (JE2 — Jump from $a = 2$)**.** *Let $n = 1 + 4q$ with q odd (so $a_t = 2$). Let $s := v_2(3q+1) \geq 1$. Then:*

1. *$T(n) = (3q+1)/2^s$ (odd positive integer).*
2. *$a_{t+1} = v_2\big((3q+1)/2^s - 1\big)$.*
3. *Height K is achieved ($a_{t+1} \geq K$) if and only if $q \equiv r_{K,s} \pmod{2^{K+s}}$, where:*

$$r_{K,s} := (2^s - 1) \cdot 3^{-1} \bmod 2^{K+s}.$$

Proof. Part (1): $3n+1 = 3(1+4q)+1 = 4(1+3q) = 4(3q+1)$. Since q is odd, $3q+1$ is even; let $s = v_2(3q+1)$. Then $v_2(3n+1) = 2+s$ and $T(n) = (3q+1)/2^s$. Since $s = v_2(3q+1)$, the quotient $(3q+1)/2^s$ is odd.

Part (2): $a_{t+1} = v_2(T(n)-1) = v_2\big((3q+1)/2^s - 1\big)$.

Part (3): Write $(3q+1)/2^s = u$ (odd). Then $a_{t+1} = v_2(u-1) \geq K$ iff $2^K \mid u-1$ iff $u \equiv 1 \pmod{2^K}$ iff $(3q+1)/2^s \equiv 1 \pmod{2^K}$ iff $3q+1 \equiv 2^s \pmod{2^{K+s}}$ iff $3q \equiv 2^s - 1 \pmod{2^{K+s}}$ iff $q \equiv (2^s-1)\cdot 3^{-1} \pmod{2^{K+s}}$. □ □

Remark 2.2 (Verification)**.** For $q = 769$ (occurring in the $n_0 = 27$ orbit at $t = 32$): $s = v_2(3\cdot 769+1) = v_2(2308) = 2$. $T(n) = 2308/4 = 577$. $577-1 = 576 = 2^6 \cdot 9$, so $a_{t+1} = 6$. Confirmed at $t = 33$ in the $n_0 = 27$ orbit.

Proposition 2.3 (JE2 target structure)**.** *The targets $r_{K,s}$ satisfy the Hensel lifting property: $r_{K',s} \equiv r_{K,s} \pmod{2^{K+s}}$ for all $K' > K$. Notable special values: $r_{K,2} = 1$ for all K (since $(2^2-1)\cdot 3^{-1} = 3\cdot 3^{-1} = 1$), and $r_{K,4} = 5$ for all K.*

3 Scope of the Cascade Phase Theory (Papers J–M)

3.1 What the phase transport model claims

Papers J–M used the identity: after a height-K jump at renewal τ_1, the odd core at any subsequent renewal τ_i satisfies $q_{\tau_i} = 3^{S_i} q_{\tau_1}$, where $S_i = \sum_{j<i} L_{K_j}$ is the cumulative cascade exponent. Under this model, Papers J–M derived waiting costs (Theorem MC), phase obstruction (Theorem PG), and universal one-shot structure (Theorem UPI).

3.2 What is exact and what fails

Proposition 3.1 (Cascade transport is exact within a cascade)**.** *Theorem OC (Paper G) states: if $a_t = k \geq 3$, then $a_{t+j} = k - 2j$ and $q_{t+j} = 3^j q_t$ for $j \leq \lfloor (k-1)/2 \rfloor$. This is an exact algebraic identity valid within the cascade (consecutive steps with $a \geq 3$).*

Proposition 3.2 (Transport fails across renewals)**.** *The identity $q_{\tau_i} = 3^{S_i} q_{\tau_1}$ fails in general when τ_1, τ_i are non-adjacent renewal events. Between renewals, T acts on the full Syracuse orbit, changing q in a complex way that depends on the residue history. Numerically: in the $n_0 = 27$ orbit, consecutive renewal values of q have ratios ranging from 0.077 to 1.545, not expressible as powers of 3.*

3.3 What remains valid

The following results from Papers J–M remain fully valid:

(a) Theorem SR (Paper I): After a height-$K \geq 4$ jump, the *immediately subsequent* jump has height $< K$. Proof uses only one cascade (Theorem OC) and one renewal (Theorem JE/JE2).

(b) Theorem VD, OC: Exact within cascades.

(c) Theorem JE, JE2: Exact at single renewal events.

(d) Theorem UPI (Paper M) within its stated scope: starting from phase L_K immediately after a K-cascade, the phase graph cannot return to $0 \pmod{2^{K-2}}$ using only the heights accessible within that cascade.

Papers J, L, M's longer-range conclusions (waiting costs, one-shot density, record chain impossibility) require the phase transport model as an additional assumption. They correctly characterize the *within-cascade* obstruction but do not directly constrain multi-record dynamics.

4 The Record Chain Problem

4.1 The two-gate target system

Definition 4.1 (Record renewal and target)**.** A *record renewal event* at time τ is a renewal event ($a_\tau \in \{1, 2\}$) at which the subsequent jump height $K_\tau = a_{\tau+1}$ exceeds all previous jump heights. By Theorem SR (exact), these form a strictly increasing sequence $K_{\tau_1} < K_{\tau_2} < \cdots$.

The *congruence target* for record i is:

$$q_{\tau_i} \equiv \begin{cases} r_{K_i} \pmod{2^{K_i}} & (\text{Gate 1, } a_{\tau_i} = 1) \\ r_{K_i, s_i} \pmod{2^{K_i + s_i}} & (\text{Gate 2, } a_{\tau_i} = 2, s_i = v_2(3q_{\tau_i} + 1)) \end{cases}$$

4.2 The inter-record transformation

Between record renewals τ_i and τ_{i+1}, the orbit evolves through $\tau_{i+1} - \tau_i$ steps of the Syracuse map T. This induces a transformation $q_{\tau_i} \mapsto q_{\tau_{i+1}}$ that is:

- *Deterministic*: fully determined by q_{τ_i} (and n_{τ_i}).
- *Complex*: depends on the full residue history, not just a multiplicative factor.
- *NOT multiplicative*: $q_{\tau_{i+1}} \neq 3^S q_{\tau_i}$ in general.

4.3 The arithmetic memory question

Open Problem 4.2 (Arithmetic memory and record compatibility)**.** Let $K_1 < K_2 < \cdots$ be a proposed infinite record sequence. The i-th record requires q_{τ_i} to lie in a specific residue class modulo 2^{K_i}.

(a) *Memory question*: Does q_{τ_i} (satisfying the K_i-record condition) influence $q_{\tau_{i+1}}$ mod $2^{K_{i+1}}$ in a systematic way? Or does the inter-record dynamics "mix" the residue class effectively?

(b) *Compatibility question*: Is there any specific initial condition n_0 and record sequence $K_1 < K_2 < \cdots \to \infty$ that can be realized simultaneously?

(c) *Obstruction question*: Do the local obstructions from Theorem SR, when applied iteratively to the record chain, accumulate into a global obstruction?

4.4 Connection to Papers A–E mixing theory

Papers A–E studied the mixing of the Markov kernel P_K (the 2-adic Haar kernel on Ω_K). The key result (Paper B, Theorem ETV) is that P_K mixes in exactly $K - 1$ steps. This means: if q_τ is close to Haar-uniform modulo 2^K, then after $K - 1$ renewal steps, q is effectively uniform modulo 2^K.

Paper N's question is the *inverse*: given that q_{τ_i} satisfies a *specific* congruence (the record condition), does the orbit's subsequent dynamics bring $q_{\tau_{i+1}}$ into general position modulo $2^{K_{i+1}}$? If the mixing is fast, then each record is "arithmetically independent" of the previous one, and infinite records are possible in principle. If there is "long-range memory", each record may constrain all future records.

5 Numerical Evidence and the Current Picture

5.1 The n0=27 example

The orbit of $n_0 = 27$ achieves records $K = 5$ (Gate 1, $t = 6$) and $K = 6$ (Gate 2, $t = 33$). The values are:

$$q_{\tau_1} = 53 \equiv r_5 = 21 \pmod{32} \quad \checkmark$$
$$q_{\tau_2} = 769 \equiv r_{6,2} = 1 \pmod{64} \quad \checkmark$$

Between $\tau_1 = 5$ and $\tau_2 = 32$, there are 27 renewal steps and the q values range widely with no simple multiplicative pattern.

5.2 Summary of proved results

Result	Status	Scope
Theorem JE: Gate 1 jump formula	Proved	Single renewal
Theorem JE2: Gate 2 jump formula	**Proved**	Single renewal
Theorem SR: immediate isolation ($K \geq 4$)	Proved	Two consecutive renewals
Theorem OC: $q \mapsto 3q$ in cascade	Proved	Single cascade
Theorem UPI: phase frozen after cascade	Proved	Within-cascade
Papers J-L-M: long-range phase theory	Conditional	Cascade-transport model
Record chain compatibility	**Open**	Multiple records
Arithmetic memory of renewals	**Open**	Long range

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